

Research Paper

Evaluation of energy and lighting in schools considering solar protections

Cristián Muñoz-Viveros^a, Carlos Rubio Bellido^b, Alexis Pérez-Fargallo^{c,*}^a Department of Theory and Design, Faculty of Architecture, Construction, and Design, University of Bío-Bío, Concepción, Chile^b Department of Building Construction II, University of Seville, Seville 41012, Spain^c Faculty of Engineering, Architecture and Design, University of San Sebastián, Concepción, Chile

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ABSTRACT

The implementation of the solar protections laid out in the regulations and certification systems decreases cooling demands, although it has an uncontrolled and even unforeseen effect on daylighting. Achieving a balance between both requirements is a challenge for facade design, energy behavior, and lighting performance since restricting solar radiation contributions decreases daylight's contribution. This research aims to define a methodology that allows, in the early design stage, the evaluation of solar protection solutions that consider the optimal performance of daylight using annual dynamic indicators while maintaining adequate energy-saving levels. For this purpose, cooling energy consumption and dynamic indicators have been considered as the primary indicators, namely Spatial daylight autonomy (sDA), Annual Sunlight Exposure (ASE), and useful daylight illuminance (UDI) with variations of the Modified Solar Factor (MSF). Thermal and light performance assessments were made using energy modeling for two types of solar protection. The case study is a school classroom located in the city of Talca, in central Chile, considering window-to-wall ratios (WWR) of 40 %, 50 %, and 60 %. The premise's thermal and light behavior were obtained with both types of solar protection. The sDA and UDI results allowed making an approximation of optimal solutions, however, the ASE values in all cases, were classified as unsuitable for classroom use. The analysis suggests that better limit values for both requirements are obtained by organizing the results by WWR instead of by the MSF that each solution identifies. This methodology compared solar protection options at an early design stage, reaching recommended light performance levels and energy savings of 70 % or more, for the two types of solar protection. It is concluded that to achieve minimum acceptable daylighting levels, in balance with cooling energy consumption, it is necessary to consider annual dynamic assessments with sDA and UDI as relevant indicators.

1. Introduction

The construction sector is responsible for most primary energy consumption, globally reaching 40 % and one-third of greenhouse gas emissions (Vaisi et al., 2023). Educational buildings account for 17 % of the non-residential building stock (Droutsas et al., 2018) and are relevant for energy savings on being large surface area buildings, holding a large number of users, and, in some cases, due to intensive use of air conditioning equipment. In the last 20 years, energy consumption in the tertiary sector has increased energy consumption by 40 % for member countries of the Organization for Economic Cooperation and Development (OECD) and by 100 % in non-OECD countries. The energy efficiency of this sector is determined by indicators that have been classified according to the end use of the energy consumed, grouped into 5 main categories: heating, cooling, water heating, lighting, and other

equipment (Energy Agency, n.d.). From this, Principal and Complementary Indicators have been defined, with the main energy indicator being Annual CO₂ Emissions (kg/m²) or Annual Primary Energy (kW/m²) and complementary, the breakdown of CO₂ emissions and primary energy consumption, and the energy demanded by the building for each of its main services. The complementary indicators explain the behavior of a building, focusing on the possible improvements (Dirección General de Arquitectura et al., 1998a; Gasparella et al., 2011; Government of Spain. et al., 2009; MINVU, 2017). Depending on the use of the building and the climate, this energy demand may be higher for heating or cooling. When efforts are focused on reducing the demand for thermal energy, other secondary energy demands or associated performances are usually unknown or disregarded. To establish a variable's behavior, energy simulation processes are required to evaluate the impact, in early design stages, that certain decisions have on end behavior and, therefore, building performance. For educational

* Corresponding author.

E-mail address: alexis.perez@uss.cl (A. Pérez-Fargallo).<https://doi.org/10.1016/j.egy.2024.07.017>

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Nomenclature and abbreviations		NCh	Chilean Standard
		N	North
		NE	Northeast
		NW	Northwest
Con	Thermal conductivity	sDA	Spatial daylight autonomy (%)
ASE	Annual Sunlight Exposure (%)	ST	Solar Transmission Factor
DB	Design Builder Software - Energy simulation and daylight analysis.	SR	Solar Reflectance Factor
DGP	Daylight Glare Probability	U	Thermal transmittance U value (W/m ² (k)
DF	Daylight factor	UDI	Useful daylight illuminance
emis	Emissivity	VT	Visible Transmission Factor
IRT	Infrared transmission factor	VR	Visible reflectance factor
kg/m ²	Kilograms per square meter	W	West
MINVU	Ministry of Housing and Urban Planning	WWR	Window-to-wall ratio (%)
MSF	Modified solar factor		

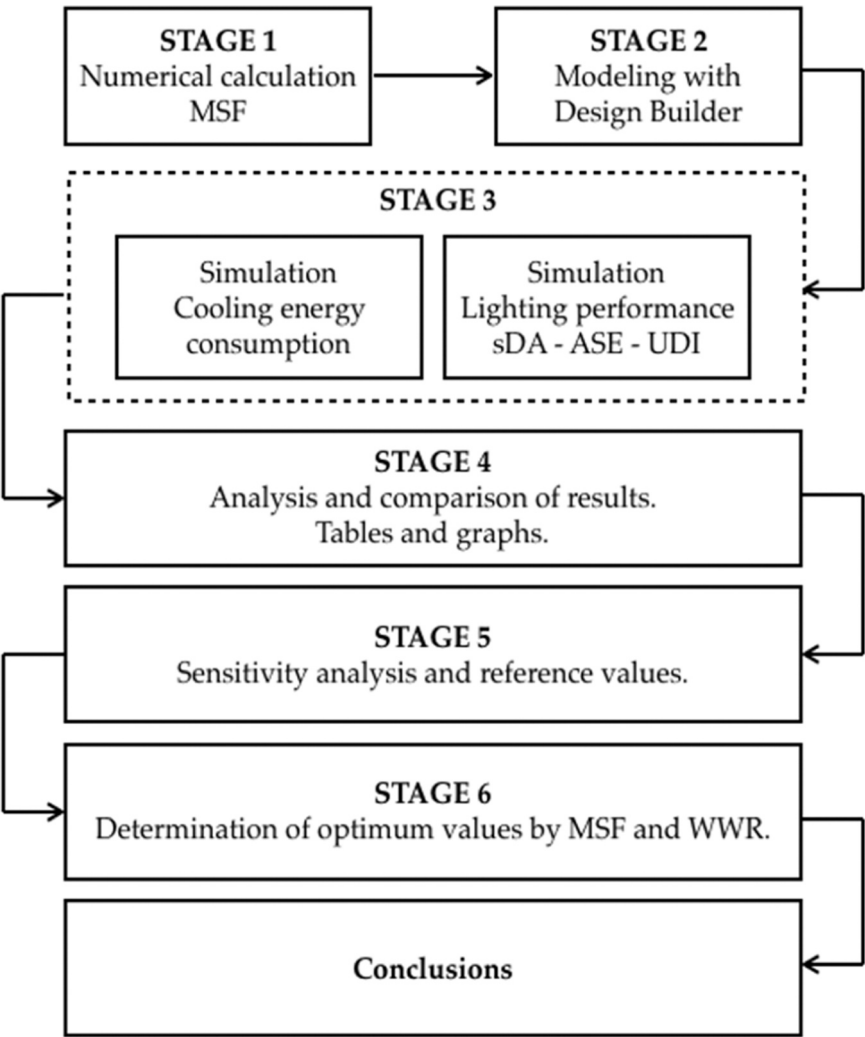


Fig. 1. Analysis process and tools.

buildings, lighting performance is important for activities that take place in the classrooms, in addition to providing psychological and physiological benefits such as improving emotions, motivation, the immune system, and the regulation of the circadian cycle, avoiding environments with artificial lighting (Bellia et al., 2015; Ma and Yang, 2023). However, this is a performance that counters the thermal, by generating opposing behaviors, where restricting the size of openings to control the entry of solar radiation, leads to a reduction of daylight contributions,

and with this, the performance of this requirement. On the other hand, providing large amounts of daylight can lead to energy savings of up to 50 % (Lo Verso et al., 2021).

Since light and thermal behavior are associated with the openings the building's envelope has, it should be considered that both the optical properties of the glazed surfaces and opaque element components of the opening, as well as the geometry of the opening, its orientation, the WWR ratio, and the characteristics of the solar control or protection

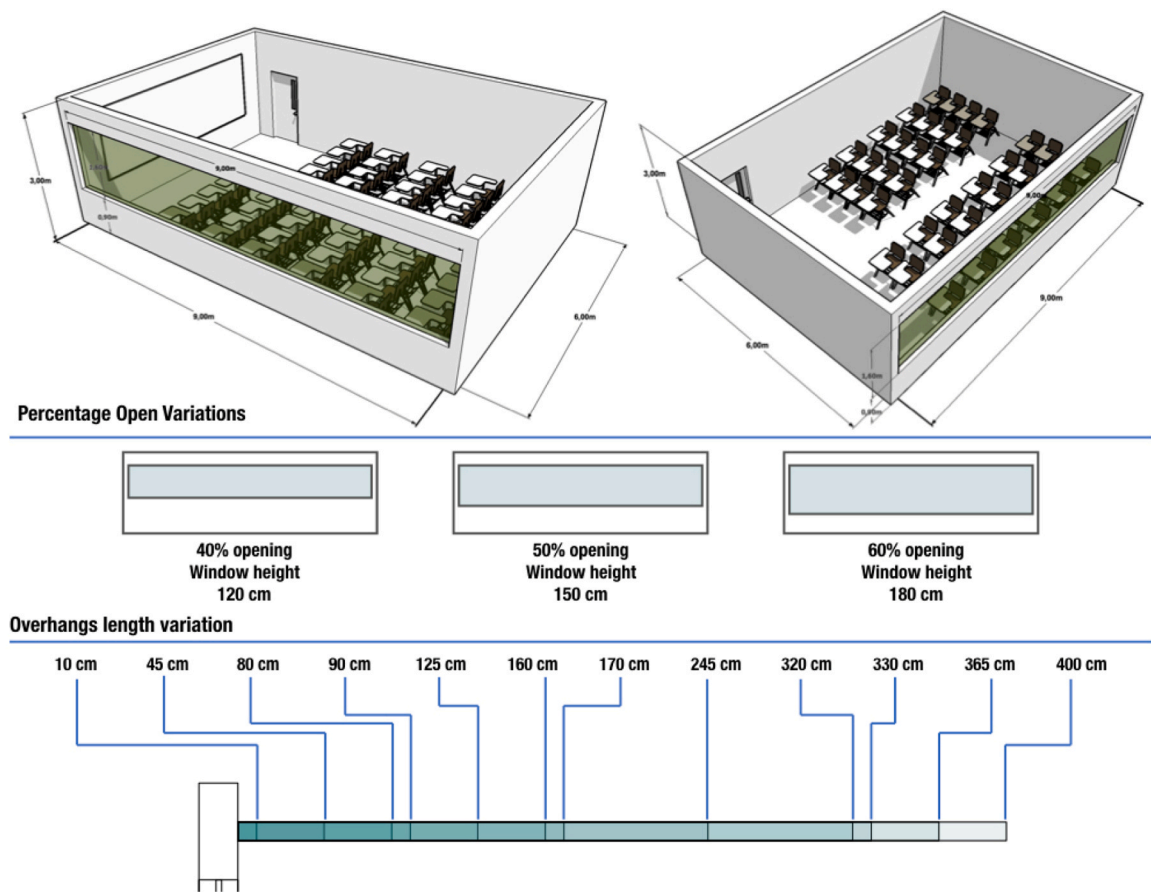


Fig. 2. Outline of the base model made with SketchUp and window-to-wall ratios (WWR) for the analyzed percentages and overhang lengths used.

Table 1
The shading factor for façade obstacles (corbel or overhang), MSF and its components, and for the setback. Source: Preparation by the Authors based on the Spanish Technical Building Code (General Direction of Architecture, 1998).

Orientation.			0.2<L/H≤0.5	0.5<L/H≤1	1<L/H≤2	L/H>2
	N	0<D/H≤0.2	0.82	0.5	0.28	0.16
		0.2<D/H≤0.5	0.87	0.64	0.39	0.22
		D/H>0.5	0.93	0.82	0.6	0.39
	NE/NW	0<D/H≤0.2	0.9	0.71	0.43	0.16
		0.2<D/H≤0.5	0.94	0.82	0.6	0.27
		D/H>0.5	0.98	0.93	0.84	0.65
	E/W	0<D/H≤0.2	0.92	0.77	0.55	0.22
		0.2<D/H≤0.5	0.96	0.86	0.7	0.43
		D/H>0.5	0.99	0.96	0.89	0.75
	N	0.05<D/H≤0.1	0.82	0.74	0.62	0.39
		0.1<D/H≤0.2	0.76	0.67	0.56	0.35
		0.2<D/H≤0.5	0.56	0.51	0.39	0.27
		D/H>0.5	0.35	0.32	0.27	0.17
	NE/NW	0.05<D/H≤0.1	0.86	0.81	0.72	0.51
		0.1<D/H≤0.2	0.79	0.74	0.66	0.47
		0.2<D/H≤0.5	0.59	0.56	0.47	0.36
		D/H>0.5	0.38	0.36	0.32	0.23
	E/W	0.05<D/H≤0.1	0.91	0.87	0.81	0.65
		0.1<D/H≤0.2	0.86	0.82	0.76	0.61
		0.2<D/H≤0.5	0.71	0.68	0.61	0.51
		D/H>0.5	0.53	0.51	0.48	0.39

elements influence its performance. (Fela et al., 2019; Zomorodian and Tahsildoost, 2017).

Although this performance can be solved with artificial lighting, as the regulatory regulations indicate, visual comfort in this type of premises prioritizes the contribution of daylighting. Certification systems such as LEED (Leadership in Energy & Environmental Design) and others guide and give additional scores for projects that consider daylighting assessments using specific parameters (De Luca et al., 2019;

Table 2
Boundary conditions of simulations: loads, setpoints, and values.

Loads	Units	Value	Schedule
Natural ventilation	Ac/h	5	Monday to Friday 8 am to 6 pm
Infiltration	Ac/h	0.7	Annual
General lighting-Normalized power density	W/m ² -100 lux	2.5	Monday to Friday 8 am to 6 pm
Occupation density	people/m ²	0.5	Monday to Friday 8 am to 6 pm
Cooling	°C	24	Monday to Friday 8 am to 6 pm
Heating	°C	21	Monday to Friday 8 am to 6 pm
Lighting control	Lux	400 work plane height 0.8 m	Monday to Friday 8 am to 6 pm

Piraei et al., 2022). These parameters have been evaluated using static methods, based on given geometric configurations and optical properties of the opening elements such as the Modified Solar Factor (MSF), and dynamic methods that combine information from climate files with energy simulation programs, obtaining annual behavior analysis, using the solar seasonal variation and the availability of daylight (Zhu et al., 2020). Among the most widely used annual dynamic assessment methods are the Daylight Factor (DF); Spatial Daylight Autonomy (sDA) (Arango-Díaz et al., 2022); Annual Sunlight Exposure (ASE) (Abdollahzadeh et al., 2020; AL-Dossary and Kim, 2020; Ekici et al., 2019, 2021; M. ElBatra and Ismaeel, 2021; Ma and Yang, 2023; Mangkuto, Fela, et al., 2019; Marzouk et al., 2022; Mostafavi et al., 2022); Daylight Glare Probability (DGP) (replacing ASE) (Mangkuto, Dewi, et al., 2019; Uribe et al., 2019); and Useful Daylight Illuminance (UDI) (Besbas et al., 2022; Montaser Koohsari and Heidari, 2022).

To maintain suitable daylighting levels, these analyses establish minimum, acceptable, and/or suggested ranges and, in some cases, lower and upper limits for their evaluations. These acceptance parameters for the different indicators do not establish a direct relationship with other performance levels such as thermal behavior. However, it can be deduced from the results whether one configuration or another can affect heating or cooling (Košir et al., 2018). Conversely, thermal evaluations of solar control strategies can suggest their light performance, which implies a synergistic formulation between both requirements so that combined compliance is easier for designers (Sepúlveda et al., 2020). Hence, it is necessary to find suitable design variables that can balance daylighting and energy performance, achieving potential energy savings by decreasing lighting and thermal energy requirements.

Table 3
Glazing used and its optical and thermal characteristics.

Glazing	ID	Thickness mm	ST	SR1	SR2	VT	VR1	VR2	IRT	emiss1	emiss2	Con W/mK
0.82*	Generic	6.000	0.820	0.074	0.074	0.900	0.081	0.081	0	0.840	0.840	1.000
0.90	14708	5.800	0.902	0.080	0.080	0.911	0.083	0.083	0	0.840	0.840	1.000
0.85	21013	6.000	0.849	0.075	0.075	0.900	0.081	0.081	0	0.840	0.840	1.000
0.72	3293	5.613	0.725	0.092	0.092	0.866	0.091	0.092	0	0.840	0.840	1.000
0.70	21547	6.360	0.703	0.074	0.072	0.840	0.082	0.081	0	0.840	0.840	0.818
0.51	5067	5.660	0.512	0.055	0.055	0.629	0.062	0.062	0	0.840	0.840	1.000
0.41	9834	5.918	0.408	0.050	0.050	0.438	0.052	0.052	0	0.840	0.840	1.000
0.27	17000	5.740	0.267	0.180	0.094	0.418	0.194	0.172	0	0.839	0.840	1.000
0.19	16122	6.107	0.189	0.254	0.256	0.240	0.279	0.266	0	0.840	0.840	0.812
0.16	17019	5.880	0.160	0.303	0.080	0.193	0.329	0.116	0	0.842	0.840	1.000
0.15	9611	5.691	0.147	0.254	0.299	0.172	0.231	0.226	0	0.840	0.773	0.985

ID: From the LBNL Window database, except for 0.82; ST: Solar Transmission Factor; SR: Solar Reflectance Factor (SR1: front; SR2: back); VT: Visible transmission factor; VR: Visible reflectance factor (VR1: front; VR2: back); IRT: infrared transmission factor; emis: emissivity (emiss1: front; emiss2: back); Con: thermal conductivity. Source: Preparation by the authors based on DesignBuilder (*) and LBNL Window data.

On the other hand, uncontrolled daylighting can cause a psychological, physiological, and biological imbalance in people (Gaviria et al., 2020); as well as overheating and glare, taking into account that the design variables with the biggest impact are the WWR, solar control devices, and windows (AL-Dossary and Kim, 2020; Salomón and Ambroggio, 2022).

The simultaneous evaluation of these and other parameters has made it possible to more effectively calibrate the contribution of both requirements, being able, through energy performance simulation processes, to establish the risks or impacts of implementing one decision or another at an early design stage and thereby determine those parameters that can provide more reliable results (Agarwal et al. n.d.). When a multi-objective evaluation is used, it is possible to address the design response in a balanced way by adjusting the result to the optimized condition and in turn achieving a more sustainable building where glare and excessive heat gains are avoided while, at the same time, minimizing energy demand, and maximizing daylight availability, and with it, visual comfort (Besbas et al., 2022; Uribe et al., 2019; Wen et al., 2023). (Wei et al., 2020) evaluate 14 Green Building certification schemes and establish 4 main components to evaluate the indoor environment quality, where thermal and lighting contribute 49 % and air quality and acoustic aspects the remaining 51 %. With this, the search to achieve the goal established by a primary indicator will be possible, as long as other performance levels, not necessarily energy-based and that ultimately establish an adequate indoor environment quality, are met.

The aim of this research is to define a methodology that allows looking closer at solar protection solutions that consider optimal daylighting performance through annual dynamic indicators, in the early design stage, while maintaining adequate energy-saving levels. For this purpose, the cooling energy consumption and the dynamic daylight lighting indicators sDA, ASE, and UDI have been considered the primary indicators using the variation of the MSF and the WWR, to come closer to a result that balances both performance levels.

2. Methodology

The research methodology was organized into 6 stages (Fig. 1). Stage 1 comprised determining the MSF by numerical calculation, using different dimensions for the depth of the overhangs and their equivalent in glazing for setback conditions, considering the established orientations and three window-to-wall ratio (WWR) levels: 40 %, 50 %, and 60 %. Stage 2 consisted of generating energy simulation and lighting performance models with Design Builder and the equivalent configuration of these solar protection variables. Stage 3 considered the results collection and simulation process associated with both solar protection strategies (overhang and setback) with the proposed WWR variations for cooling energy consumption (kW/hour per year; kW/hour m² per year and percentage savings (%) compared to the highest value of the series) and annual lighting performance (sDA, ASE, and UDI). Stage 4 included

Table 4
MSF values by orientation for overhangs between 0.1 and 4.0 m and glazed surfaces considering the WWRof 40 %, 50 %, and 60 %.

PROT N/E/W/ NE/NW	HW = 120 WWR 40 %	HW = 150 WWR 50 %	HW = 180 WWR 60 %
OV 10 N	0.54	0.54	0.54
OV 45 N	0.54	0.54	0.54
OV 80 N	0.33	0.33	0.54
OV 90 N	0.33	0.33	0.54
OV 125 N	0.18	0.33	0.33
OV 160 N	0.18	0.18	0.33
OV 170 N	0.18	0.18	0.33
OV 245 N	0.11	0.18	0.18
OV 320 N	0.11	0.11	0.18
OV 330 N	0.11	0.11	0.18
OV 365 N	0.11	0.11	0.11
OV 400 N	0.11	0.11	0.11
SF 0.85 N	0.54	0.54	0.54
SF 0.51 N	0.33	0.33	0.33
SF 0.27 N	0.18	0.18	0.18
SF 0.16 N	0.11	0.11	0.11
OV 10 E/W	0.61	0.61	0.61
OV 45 E/W	0.61	0.61	0.61
OV 80 E/W	0.51	0.51	0.61
OV 90 E/W	0.51	0.51	0.61
OV 125 E/W	0.36	0.51	0.51
OV 160 E/W	0.36	0.36	0.51
OV 170 E/W	0.36	0.36	0.51
OV 245 E/W	0.14	0.36	0.36
OV 320 E/W	0.14	0.14	0.36
OV 330 E/W	0.14	0.14	0.36
OV 365 E/W	0.14	0.14	0.14
OV 400 E/W	0.14	0.14	0.14
SF 0.85 E/W	0.60	0.60	0.60
SF 0.51 E/W	0.51	0.51	0.51
SF 0.27 E/W	0.36	0.36	0.36
SF 0.16 E/W	0.14	0.14	0.14
OV 10 NE/NW	0.59	0.59	0.59
OV 45 NE/NW	0.59	0.59	0.59
OV 80 NE/NW	0.47	0.47	0.59
OV 90 NE/NW	0.47	0.47	0.59
OV 125 NE/NW	0.28	0.47	0.47
OV 160 NE/NW	0.28	0.28	0.47
OV 170 NE/NW	0.28	0.28	0.47
OV 245 NE/NW	0.11	0.28	0.28
OV 320 NE/NW	0.11	0.11	0.28
OV 330 NE/NW	0.11	0.11	0.28
OV 365 NE/NW	0.11	0.11	0.11
OV 400 NE/NW	0.11	0.11	0.11
SF 0.85 NE/NW	0.60	0.60	0.60
SF 0.51 NE/NW	0.47	0.47	0.47
SF 0.27 NE/NW	0.28	0.28	0.28
SF 0.16 NE/NW	0.11	0.11	0.11

Note: PROT: Type of solar protection; HW: Height of window opening; WWR: window-wall percentage; OV: Overhangs; SF: Glazing shading factor.

the analysis and comparison of the results of the 4 indicators. Stage 5 consisted of the sensitivity analysis of the variables using the R Studio software, to determine the impact of the parameters used, and establish reference values for the evaluated dimensions. Stage 6 consisted of determining optimal MSF and WWR values for each orientation.

The threshold values for sDA and UDI consider 50 % as the lower limit; values equal to or greater than 50 % are considered acceptable and for ASE sets a maximum value of 10 %; acceptable, 7 %; and recommended, 3 % or less. For cooling energy consumption, the 40th percentile value has been used as a reference. This value is defined in the document "Energy Efficiency Rating of Buildings 7, Energy Rating Scale, New Buildings, Point 3.2 Obtaining the Scale Limits," indicating that the boundary between classes C and D corresponds to the 40th percentile for buildings that meet the requirements of the CTE. Although it does not align with the proposal by the CEN (European Committee for Standardization) to place it between classes B and C, this reference encourages the improvement of the thermal quality of buildings (Gobierno de

España. et al., 2009).

The experiment was made in the city of Talca, Chile, which has a marked seasonal variability. The cooling energy consumption and daylighting performance have been analyzed in a standard school classroom with WWR variations, considering solar protections with overhangs and optimized glazing for one year.

2.1. Base case

A 9.00 × 6.00 × 3.00 m school classroom was used, with 20 cm thick reinforced concrete walls and a glazed surface on one of its facades with a WWR of 40 %, 50 %, and 60 % equivalent to a continuous 9 m long and 1.2 m, 1.5 m, and 1.8 m high pane, maintaining a lintel height at 2.40 m from the indoor floor level. The glazed facade is in contact with the outside and determines the model's orientation. The other facades of the model have been considered adiabatic. The characteristics and dimensions of the model are shown in Fig. 2.

The formula established in the Basic HE Document of the Spanish Technical Building Code was used to determine the MSF and the tables for the sizes of the openings and solar protection elements. The table established for the overhang and setback elements has also been used (Table 1)(Dirección General de Arquitectura et al., 1998b). The MSF can be described as follows:

$$MSF = SF \cdot [(1 - F_v) \cdot g_{\perp} + FF \cdot 0,04 \cdot UF \cdot \alpha]$$
 (1)

Where MSF is the Modified Solar Factor, SF is the shading factor, F_v is the fraction of the gap occupied by the glass, g_{\perp} is the solar factor of the glass (UNE EN 410:1988), FF is the fraction of the gap occupied by the frame, UF is the transmittance of the frame (W/m^2(K) and \alpha is the absorbance, considering the color of the window frame. To obtain the value for the geometry of the case study's glazed surface, solar protection variants were used with overhangs from 10 to 400 cm deep, single glazing, and half-white aluminum frames with a thermal bridge break for the North, Northeast, Northwest, East, and West orientations. To determine the characteristics of the case study's envelope, those defined in the Energy Efficiency Guide for Educational Establishments of the Chilean Energy Efficiency Agency (2012) were used.

The dynamic light performance indicators used are Special Daylight Autonomy (sDA), Annual Sunlight Exposure (ASE), and Useful Daylight Illuminance (UDI). sDA and ASE are part of the Lighting Measurement 83 (LM-83) guide published in 2013 by the Illuminating Engineering Society (IES). sDA is derived from the Daylight Autonomy (DA) indicator. It is defined as the percentage of the space's area that has an illuminance of at least 300 lux for at least 50 % of the occupied hours during a year, considering a daily occupancy from 8:00 am to 6:00 pm. It sets a minimum value of 50 %, with suggested ranges where 55 % is Acceptable and 75 % is Preferred. ASE was conceived to reduce the risks of glare and overheating associated with the intensive use of daylighting in buildings. It is considered a complement to sDA. ASE looks to prevent spaces from receiving an excessive amount of solar radiation. It is defined as the percentage of the space that has an illuminance of 1000 lux or higher for 250 hours a year or more, considering a daily occupancy from 8:00 am to 6:00 pm. An ASE value of 25 % indicates that a quarter of the space exceeds the established conditions. Values above 10 % can lead to occupant visual discomfort, with 10 % being understood as Sufficient, 7 % as Acceptable, and 3 % as Preferred. UDI establishes the illuminance on the work plane within ranges considered suitable for performing normal activities without generating visual discomfort. It considers a minimum annual value of 50 % of the time within the illuminance ranges of 300–3000 lux. The results obtained by simulation indicate the area of an enclosure that is kept within those ranges (Ayoub, 2019).

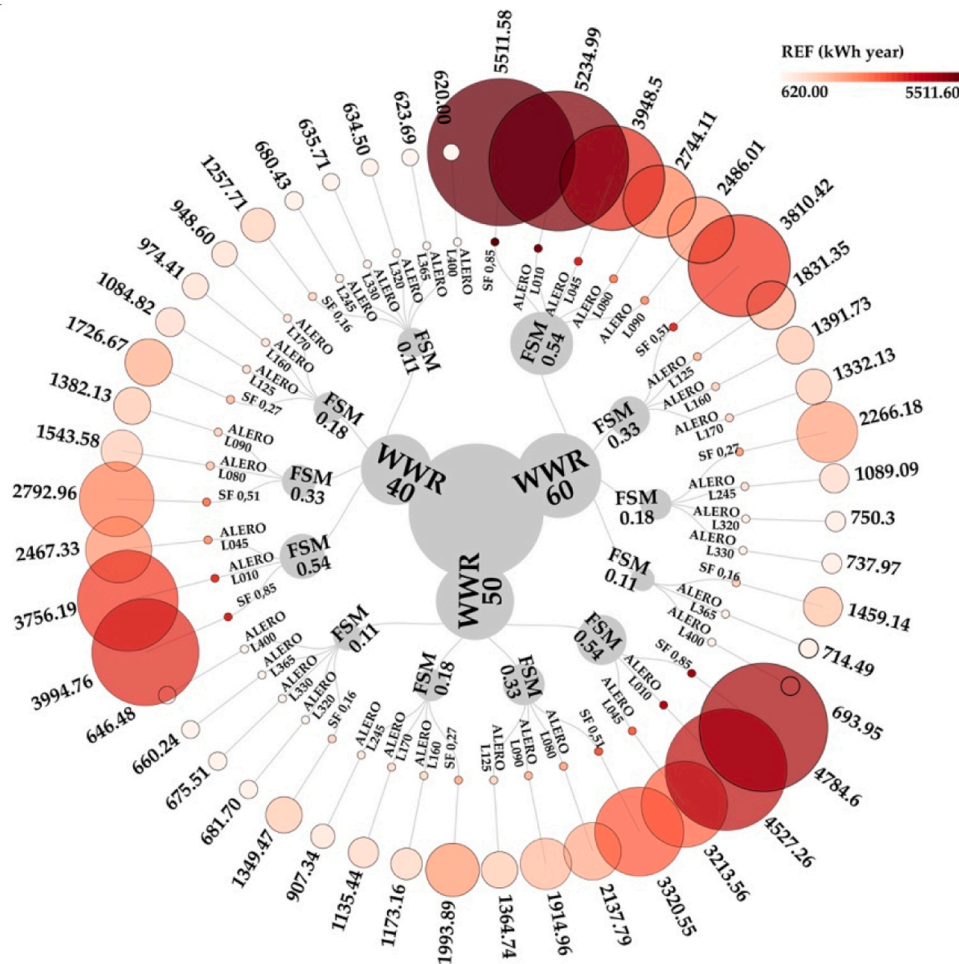


Fig. 3. Cooling energy consumption values by WWR and type of solar protection for the North orientation.

2.2. Boundary conditions and simulation

The climate file of the location under evaluation was obtained using the Meteonom v.8.0 software, considering the official station this suggested. To determine the daily solar radiation ($\text{Wh/m}^2 \times \text{day}$) at the site, data was taken from the Chilean Standard 1079 Of 2019 and the Climate Consultant software v 6.0, as well as the EPW file obtained, for those values that were not found in the Standard's table. For the city of Talca, the qualitative and appreciation values, according to NCh 1079, are an average temperature of 20.3°C (very high - very hot) in January, and 7.6°C (low - cold) in July. The average monthly oscillation in January is 18.9°C (average), and in July, 11.4°C (average). The sunshine in January is $6445 \text{ Wh m}^2 \text{ day}$ (Strong), and in July, $926 \text{ Wh m}^2 \times \text{day}$ (Very low). Classroom occupancy periods are set as Monday to Friday, from 8 a.m. to 6 p.m., with an occupancy of 27 people, and an infiltration of 0.7 air renewals per hour (ac/h), for one year, without discounting winter and summer holidays (Table 2). Lighting systems are considered as being on when 400 lux is required at 0.8 m above the work plane.

For the adiabatic envelope of the classroom's walls, reinforced concrete floors and roofs are used without coating, except for the wall containing the glazed surface which has a 60 mm thick expanded polystyrene thermal insulation with the U value of $0.59 \text{ W/m}^2\text{K}$ established by local regulations.

The Design-Builder (DB) energy simulation program uses the EnergyPlus calculation engine, which makes it possible to model geometries of complex openings and efficiently incorporate boundary conditions and the modification of the overhang and glazing parameters, covering,

in these dimensions, the entire range of values that can be obtained by applying the numerical relationships indicated in Table 1.

The MSF values for glazed surfaces are defined using calculations for the overhang and its equivalent, a 0.1 m deep setback. The values for the overhangs are separated into 4 ranges: 0.10–0.45–0.80 m; 0.90–1.25–1.60 m; 1.70–2.45–3.20 m; and, 3.30–3.65–4.00 m. These ranges establish varying values for the N, E-W, and NE-NW orientations. In turn, each of these conditions is evaluated with the WWR values of 40–50 and 60 %.

For the simulations with overhang, single glazing is established, with an ST of 0.82. The optical glazing characteristics are presented in Table 3.

With the MSF values of the combinations determined through the calculation process, these are input in the DB model to obtain the energy consumption values for cooling and annual lighting performance. This is repeated for each MSF and ST combination defined. The overhang condition is evaluated with an ST of 0.82 and the overhang sizes established for the orientations (N, NE, NW, E, and W). This allows making a comparison of the values obtained and, subsequently, determining the impact that the use of one strategy or another has on the energy consumption for cooling and the lighting performance using the sDA, ASE, and UDI indicators.

The R Studio software was used to determine the statistical values associated with the results obtained considering the total cooling energy data and their dependence on the sDA, ASE, and UDI values. Once the energy simulation results were obtained, the data were entered into RStudio to run a sensitivity analysis, considering daylighting and cooling energy consumption indicators. A p-value of 0.05 was considered as

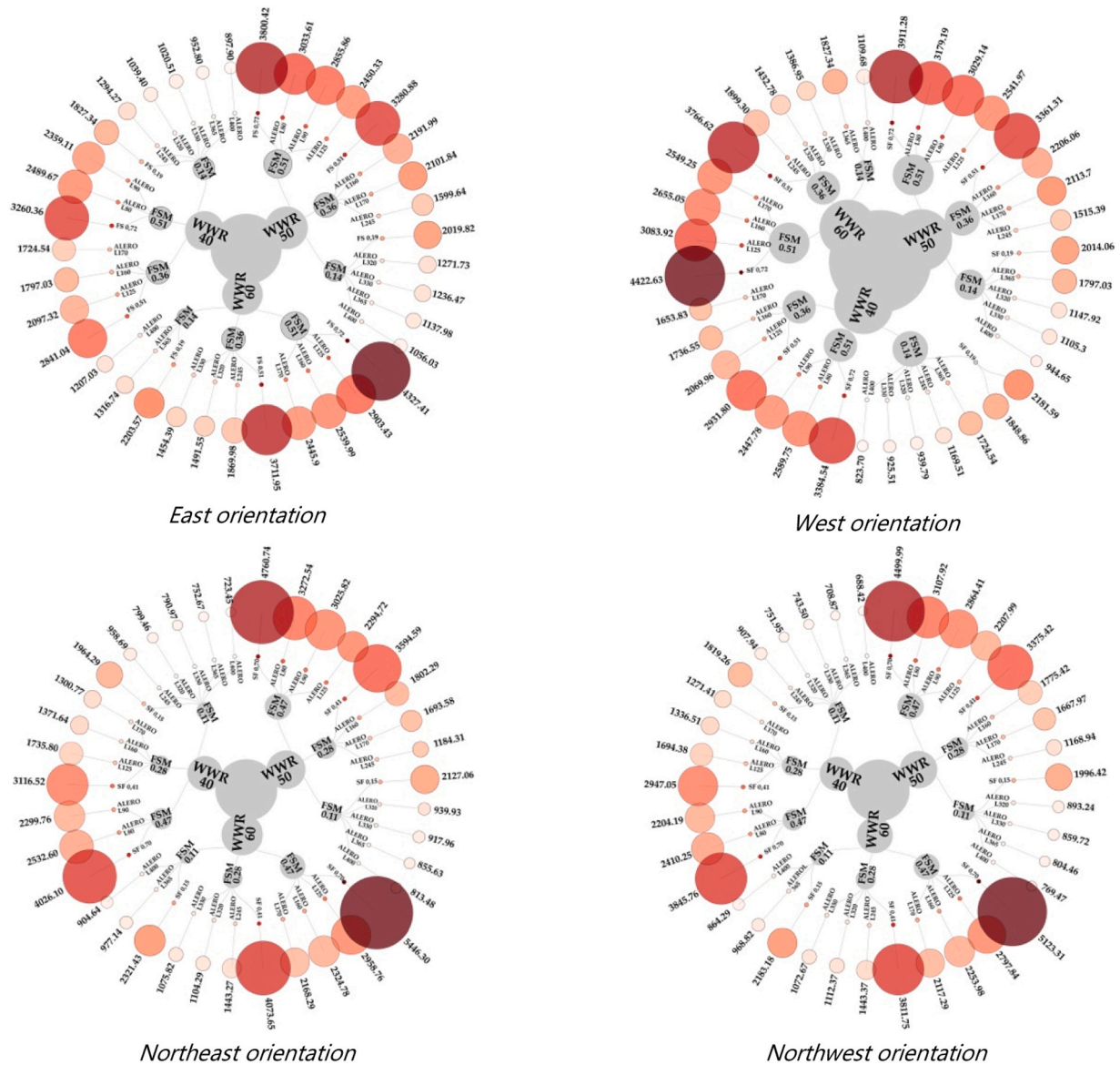


Fig. 4. Cooling energy consumption values by WWR cooling and type of solar protection for East, West, Northeast, and Northwest orientations.

an indicator of significance for the variables.

Stage 6 considers the results analysis process using the data obtained from the energy simulations. Performance was assessed through simulations considering different types of solar protection (overhang and glazing), Window-to-Wall Ratios (WWR) of 40 %, 50 %, and 60 %, and the Solar Factor (SF) of the glazing. The analysis included tabulating and organizing data to identify configurations that provide optimal thermal conditions.

The optimal condition in this study includes energy reduction and visual comfort. Here are the criteria used to define these optimal conditions:

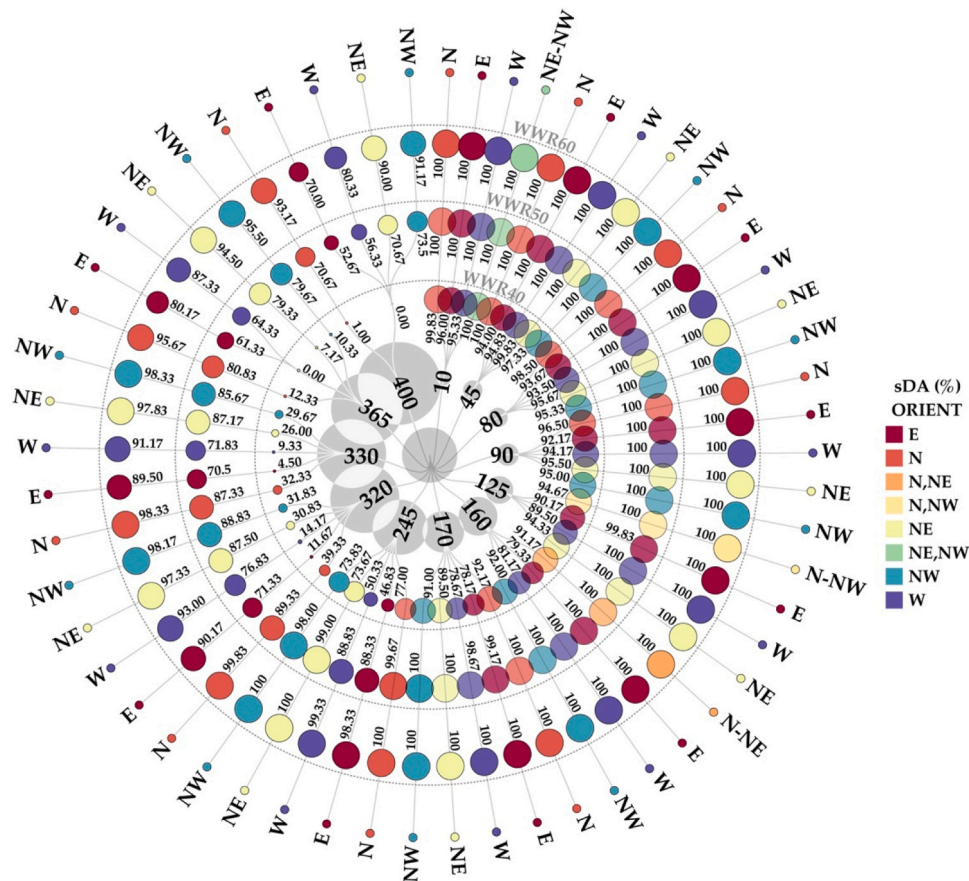
- Energy Reduction: cooling energy consumption (kWh/m² per year) and percentage energy savings. The energy-saving variable was ordered from highest to lowest, and a restriction line was set at the value of the 40th percentile or the closest one to that. This approach helps in identifying configurations that offer significant energy savings.
- Visual Comfort: Visual comfort was measured using annual dynamic daylighting indicators, namely Spatial Daylight Autonomy (sDA) and Useful Daylight Illuminance (UDI), with minimum values of 50 % for

both sDA and UDI. These indicators ensure that the design provides sufficient natural light without causing glare or overheating.

For each orientation and WWR, the following combinations of optimal values were identified:

- Lowest cooling energy consumption (green line to indicate the optimal energy solution).
- Best performance in Spatial Daylight Autonomy (sDA) and Useful Daylight Illuminance (UDI) (blue line to denote the optimal lighting solution).
- Reference based on the 40th percentile of the energy consumption (red line).

This comprehensive analysis allows us to balance both energy efficiency and occupant comfort effectively. We believe that these criteria provide a robust framework for defining and achieving the 'optimal' condition in building design.



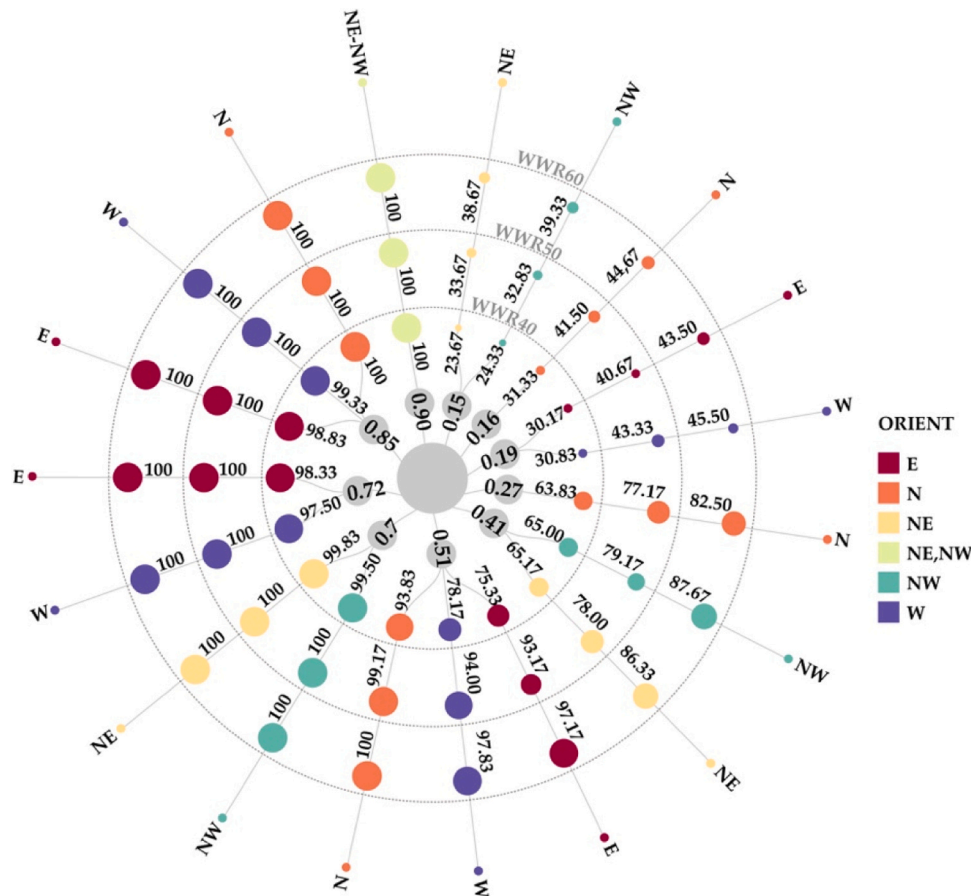


Fig. 6. SDA values for the glazing and WWR used in the N, E-W, and NE-NW orientations.

maximum (10 %), acceptable (7 %), and preferred (3 %) values, with the deeper overhangs in all the WWR having the highest values. The lowest values (best results) are for the WWR50 and WWR60 for the overhang lengths of 10 cm and 45 cm.

It is seen that the glazing does not reach suitable values in any of the series, with the highest values being associated with SF 0.16, 0.19, and 0.15 under WWR60 and the lowest under SF 0.85, 0.90, 0.51, 0.72, and 0.70 with WWR50 and WWR60.

3.3.3. UDI results

A minimum value of 50 % of the time within the illuminance range of 300–3000 lux is established. Fig. 9 summarizes the values obtained for the different overhang sizes and WWR values, and Fig. 10 summarizes the values for the glazing.

It is seen that all the data obtained for overhangs exceed the recommended value. The values with WWR40 and lengths up to 330 cm reach the maximum, as does WWR50–60, with lengths up to 400 cm.

It is seen that all the glazing reaches the recommended value, with SF 0.27, SF 0.51, and SF 0.41 having a better performance.

3.4. Definition of optimal value. Grouping of data by WWR

The calculation of the MSF allows linking different types of solar protection and WWR under the same value. This implies that, at the same MSF value, there should be the same thermal performance, which has already been proven not to be the case, not only between different types of solar protection but also between variants of the same type, attaining significant differences in the results (Muñoz-Viveros et al., 2022). With this background information, it is possible to assume that the lighting performance may have variations of the same type.

The results obtained through the different simulations have been

grouped by orientation and by WWR. The data are organized following the cooling energy savings considering the highest value of the analyzed data series.

Since all the results obtained for the ASE variable remain outside the range, these have been discarded from the following analyses.

The data have been grouped according to the WWR that characterizes them, independent of their MSF. This criterion allows making a similar approach to what the regulatory documents establish when identifying a design parameter as a starting point to define the type of solar protection.

The significance of the parameters is evaluated for each series of values, using RStudio to establish whether all those included need to be considered.

For WWR40–50 and 60 the value of R², related to the effectiveness of the model, was over 87 %, which indicates that the variables as a whole explain the result of cooling Energy Consumption (Table 5). The p-value (statistical contrast) obtained was less than 0.05, which is why it is considered valid with a significance of the variables above 99.9 %, with the UDI value having the greatest significance for the daylighting indicators in the three cases.

3.5. Definition of the optimized cooling energy-light performance values

Graphs were built, based on the results obtained, with the 0.1–0.9 percentiles for the variables for each WWR defined.

3.5.1. North orientation

Table 6 indicates the 40th percentile values for the North orientation.

A graph was made, with the data obtained and considering ordering the energy-saving variable from highest to lowest, establishing the restriction lines for the different variables. For SDA and UDI, a straight line

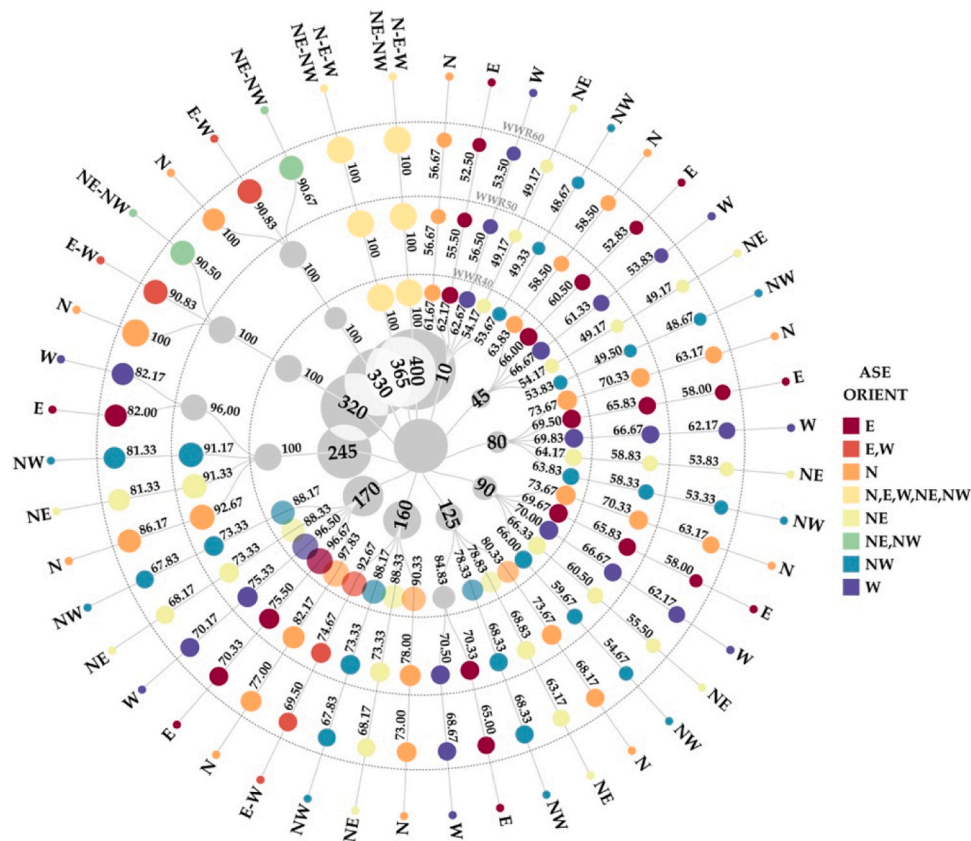


Fig. 7. ASE values for the overhang sizes and WWR used, N, E-W, and NE-NW.

is set at 50 %, and for cooling energy consumption, at the value of the 40th percentile or the closest one to that.

3.5.1.1. North orientation WWR40. Fig. 11 shows, in the left column, the graphs associated with the 10th to 90th percentiles of each of the North orientation WWRs. In the right column, the graphs associated with the results of each variable of each series (1–16: twelve overhang length: 10,45,80,90,125,160,170,245,320,330,365,400 cm and four MSF: 0,11;0,18;0,33; 0,54) are presented by WWR and their limit values.

The results determine the following results for the solar protection by WWR. These are shown in Fig. 12.

The solution of SF 0.51 and MSF 0.33 is the lower limit for cooling energy performance, establishing 9 solutions ranging between 12.13 kWh/m² year to 49.79 kWh/m² year, the equivalent of 82.97 % and 30.08 % savings compared to the highest value in the series. The overhang solution of 80 cm and MSF 0.33 has the best light performance (sDA = 98.50 % and UDI = 100 %) and is within the best thermal performance series with 27.51 kWh/m² year and 61.36 % savings.

For WWR40, the upper limit is with a 245 cm deep overhang and MSF 0.11, with a maximum energy consumption saving value equivalent to 82.97 % (680.43 kWh/m² year or 12.13 kWh/m² year), with sDA reaching 77 %, and UDI, 100 %. The lower limit is defined by the 80 cm deep overhang and MSF 0.33, with a cooling energy consumption level that has a 61.36 % energy saving (1543.58 kWh/m² year or 27.51 kWh/m² year) compared to the highest value of the series. In this case, it coincides with the point where the sDA and UDI reach maximum performance. After that value, the daylight performance values remain high, but the thermal performance is considerably reduced. Fig. 13

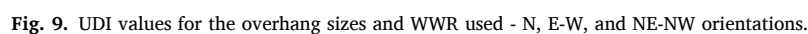
3.5.1.2. North orientation WWR50. The 90 cm overhang solution and MSF 0.33 is the lower limit for the cooling energy performance, establishing 9 solutions ranging between 11.52 kWh/m² year to 34.13 kWh/m² year,

the equivalent of 86.49 % and 59.98 % savings compared to the highest value in the series. The 170 cm overhang solution and MSF 0.18 have the best light performance (sDA and UDI = 100 %). This is within the best thermal performance series with 20.24 kWh/m² year and 76.27 % savings.

For WWR50, the upper limit is with a 365 cm deep overhang with an MSF of 0.11, an energy consumption saving value equivalent to 86.20 % (660.24 kWh/m² year or 11.77 kWh/m² year), and where sDA reaches 80.83 % and UDI, 100 %. The 170 cm deep overhang with an MSF of 0.18 coincides with the point where sDA and UDI reach their maximum performance with energy savings of 76.27 % (1135.44 kWh/m² year or 20.24 kWh/m² year). The lower limit is defined by the 125 cm deep overhang and an MSF of 0.33, with a cooling energy consumption level that has a 71.48 % energy saving (1364.74 kWh/m² year or 24.33 kWh/m² year) compared to the highest value in the series. After that value, the daylight performance values remain high, but the thermal performance is considerably reduced. Fig. 14

3.5.1.3. North orientation WWR60. The 160 cm overhang solution and an MSF 0.33 is the lower limit for the cooling energy performance, establishing 7 solutions ranging between 12.37 kWh/m² year to 24.81 kWh/m² year, the equivalent of 87.41 % and 74.75 % savings compared to the highest value in the series. The 245 cm overhang solution and an MSF of 0.18 has the best light performance (sDA and UDI = 100 %). This is within the best thermal performance series with 19.41 kWh/m² year and 80.24 % savings.

For WWR60, the upper limit is with a 400 cm deep overhang with an MSF of 0.11, an energy consumption saving value equivalent to 87.41 % (693.95 kWh/m² year or 12.37 kWh/m² year), and where sDA reaches 93.17 %, and UDI, 100 %. The 245 cm deep overhang and an MSF of 0.18 coincides with the point where sDA and UDI reach their maximum performance (100 %) with the highest energy savings of 80.24 %



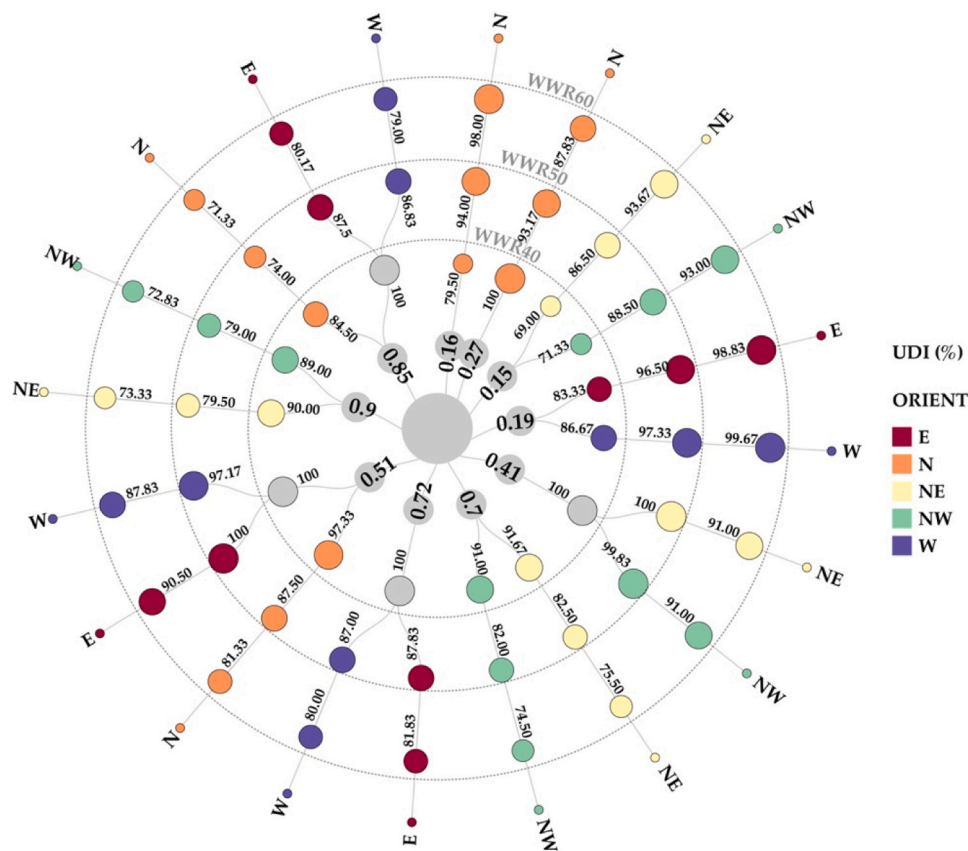


Fig. 10. UDI values for the glazing and WWR used. N, E, W, and NE-NW orientation.

Table 5
Values for regression model considering: Cooling energy, WWR, sDA, ASE, and UDI for overhangs and glazing with a given MSF.

NORTH WWR40 without ASE					
Parameter	Estimation	Standard Error	t value	Pr(> t)	Significance
Intercept	4592.752	1993.589	2.304	0.03992	*
MSF	4927.274	1163.775	4.234	0.00116	**
sDA	1.498	4.826	0.310	0.76155	
UDI	-45.307	20.122	-2.252	0.04387	*
Residual standard error: 439.7 on 12 degrees of freedom					
Multiple R-squared: 0.8741, Adjusted R-squared: 0.8427					
F-statistic: 27.78 on 3 and 12 DF, p-value: 1.102e-05					
NORTH WWR50 without ASE					
Parameter	Estimation	Standard Error	t value	Pr(> t)	Significance
Intercept	14918.900	4266.219	3.497	0.00441	**
MSF	1896.759	2365.449	0.802	0.43823	
sDA	-3.369	10.665	-0.316	0.75754	
UDI	-143.984	43.179	-3.335	0.00595	**
Residual standard error: 491.8 on 12 degrees of freedom					
Multiple R-squared: 0.9161, Adjusted R-squared: 0.8951					
F-statistic: 43.65 on 3 and 12 DF, p-value: 9.886e-07					
NORTH WWR60 without ASE					
Parameter	Estimation	Standard Error	t value	Pr(> t)	Significance
Intercept	20913.655	4422.362	4.729	0.000489	***
MSF	-3934.061	2910.735	-1.352	0.201442	
sDA	-4.111	13.740	-0.299	0.769911	
UDI	-193.938	43.584	-4.450	0.000793	***
Residual standard error: 636 on 12 degrees of freedom					
Multiple R-squared: 0.873, Adjusted R-squared: 0.8412					
F-statistic: 27.49 on 3 and 12 DF, p-value: 1.163e-05					

(1089.09 kWh/m² year or 19.41 kWh/m² year). The lower limit is defined by the 125 cm deep overhang with an MSF of 0.33, and a cooling energy consumption level of 66.77 % energy saving (1831.35 kWh/m² year or 32.64 kWh/m² year) compared to the highest value in the series. After that value, the daylight performance values remain high, but the thermal performance is considerably reduced.

3.5.2. East-west and northeast-northwest orientations

The same procedure was applied to the other orientations, obtaining the results summarized in the following tables.

Table 7 indicates the 40th percentile values for East and West orientation.

Figs. 15 and 16 represent the characteristic values of the solutions with WWR40–50 and 60 in East and West orientations and optimal solutions for the cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI. The green mark indicates the optimal energy solution, the blue one, the optimal light, and the red one, the energy consumption reference based on the 40th percentile.

3.5.3. Northeast-northwest orientation

Table 8 indicates the 40th percentile values for the Northeast and Northwest orientations.

Figs. 17 and 18 represent the characteristic values of the solutions with WWR40–50 and 60 in Northeast and Northwest orientations and optimal solutions for the cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI. The green mark indicates the optimal energy solution, the blue one, the optimal light, and the gray one, the energy consumption reference based on the 40th percentile.

Table 6
40th percentile values for the parameters of the different MSFs applied to the North orientation.

WWR NORTH	PERCENTILE	REF kWh year	REF_m2 kWh/m ² year	REFporc_ah %	sDA %	UDI %
40	40	974.41	75.61	65.40	77.00	99.83
50	40	1.173	59.98	34.14	99.17	93.17
60	40	74.75	1391.73	58.88	100.00	84.00

WWR: Window-to-wall ratio; REF: Energy consumption for cooling REF_m2: energy consumption for cooling per square meter; REFporc_ah: percentage of cooling energy savings based on the highest value in the series; sDA: spatial light autonomy; UDI: Useful daylight illuminance.



Fig. 11. Results of each solution (1st to 16th) and percentiles (10th to 90th) for cooling energy consumption parameters in kWh/m² year; percentage of energy savings, sDA, and UDI for WWR40–50 and 60 considering the North orientation MSF 0.11–0.18–0.33–0.54.

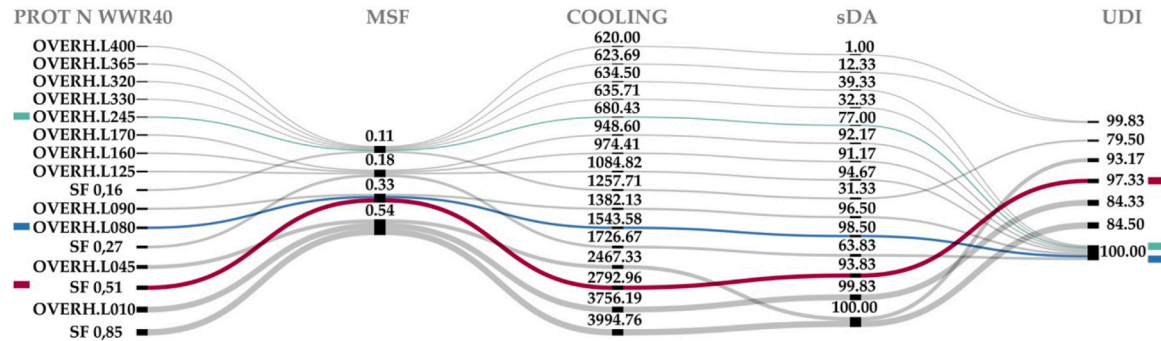


Fig. 12. Characteristic values of the North orientation WWR40 solutions and optimal solutions for cooling energy consumption parameters in kWh/m² year; percentage of energy savings, sDA, and UDI. The green mark indicates the optimal energy solution, the blue one, the optimal light, and the red one, the energy consumption reference based on the 40th percentile.

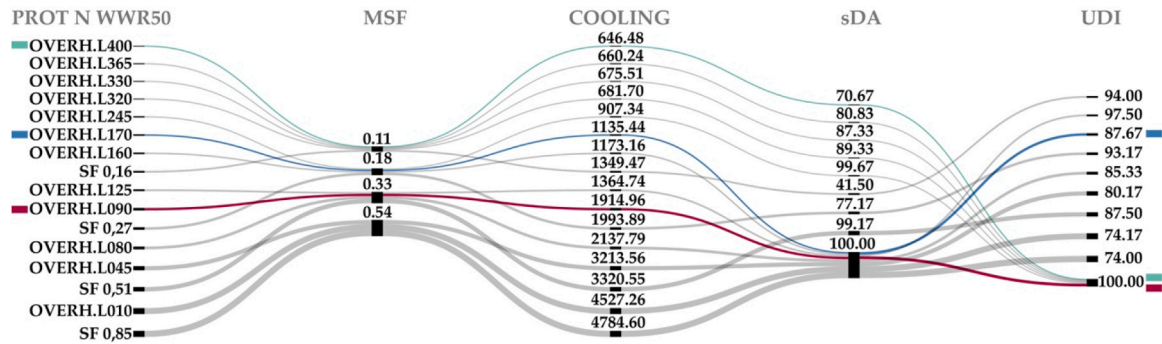


Fig. 13. Characteristic values of the North orientation WWR50 solutions and optimal solutions for cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI. The green mark indicates the optimal energy solution, the blue one, the optimal light, and the red one, the energy consumption reference based on the 40th percentile.

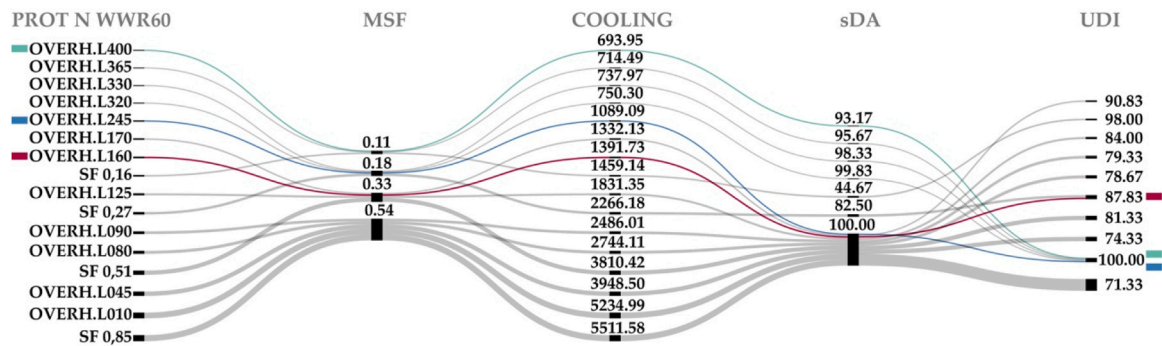


Fig. 14. Characteristic values of the North facing WWR60 solutions and optimal solutions for cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI. The green mark indicates the optimal energy solution, the blue one, the optimal light, and the red one, the energy consumption reference based on the 40th percentile.

Table 7

40th percentile values for the parameters of the different WWRs applied to the East and West orientations.

WWR EAST-WEST %	PERCENTILE	REF kWh year	REF m ² kWh/m ² year	REFporc_ah %	sDA %	UDI %
40 E	40	1638.49	29.25	49.75	43.50	99.43
50 E	40	1935.78	34.56	49.06	84.93	100.00
60 E	40	1869.98	33.39	56.79	90.17	98.83
40 W	40	1726.94	47.40	48.98	46.47	99.74
50 W	40	1970.66	35.18	49.62	86.43	99.24
60 W	40	1870.52	33.40	57.71	95.90	97.47

WWR: Window-to-wall ratio; REF: Energy consumption for cooling REF_m2: energy consumption for cooling per square meter; REFporc_ah: percentage of cooling energy savings based on the highest value in the series; sDA: spatial light autonomy; UDI: Useful daylight illuminance.

4. Discussion

The optimal cases for the MSF and WWR obtained from the results are indicated in Table 9.

4.1. Results organized by WWR

The energy consumption makes it possible to sort the data and thereby contrast the behavior of both light performance curves. It was observed that when the light performance curves tend to stabilize at the maximum values, the energy saving decreases suggesting excess solar radiation.

4.2. North orientation

When grouped by WWR and prioritizing energy or light performance, the differences observed for WWR 40 for energy performance vary by +126 %; the sDA, 21.50 %, and the UDI, 0 % (both with 100 %),

with a difference of 863.15 kWh year or 15.38 kWh/m² year. For WWR 50, the energy performance varies by +75.63 %; the sDA, 29.33 %, and the UDI, 0 % (both with 100 %), with a difference of 486.96 kWh year or 8.72 kWh/m² year and for WWR 60 the energy performance varies by 56.94 %; the sDA, 6.83 %, and the UDI, 0 % (both with 100 %), with a difference of 405.14 kWh year or 7.04 kWh/m² year.

For all the WWRs considered, the best energy performance values have a variation of less than 1 kWh/m² year, and between the light performances, this variation reaches between 7 and 9 kWh/m² year. The differences are greater between the optimal ones of each performance by WWR exceeding in some cases twice the lower value associated with the energy performance of all cases. It is observed that for WWR40, 50, and 60, the optimal thermal performance is associated with MSF 0.11 (Overhang of 245 and 400 cm), and with lighting for WWR50–60, an MSF of 0.18 (Overhang of 170 and 245 cm) and for WWR40, an MSF of 0.33 (Overhang of 80 cm).

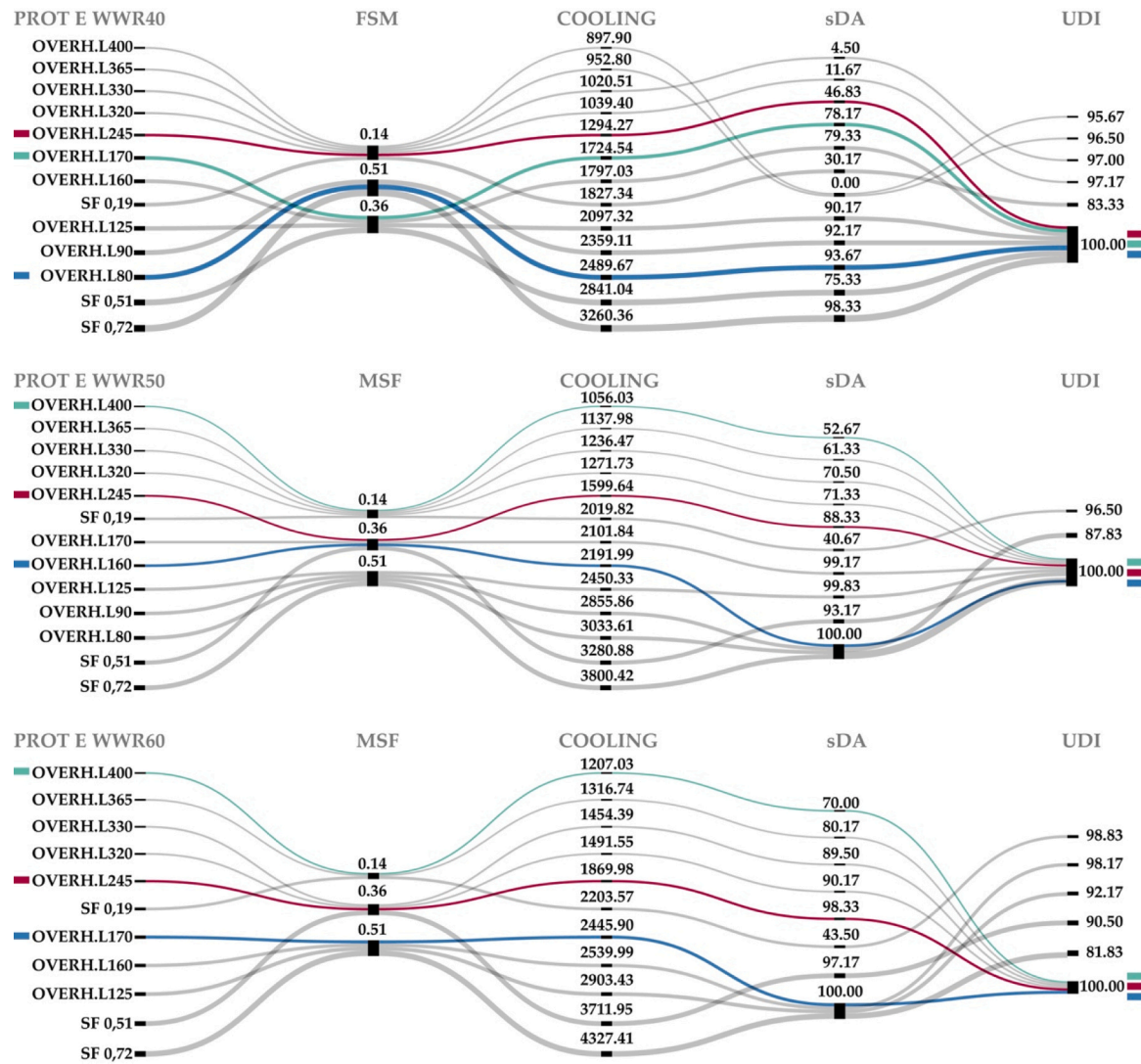


Fig. 15. Characteristic values of the East facing solutions with WWR40–50 and 60 and optimal solutions for the cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI.

4.3. East orientation

For WWR 40, the energy performance varies by +44.37 %; the sDA, 15.50 %, and the UDI, 0 % (both with 100 %), with a difference of 765.13 kWh year or 13.66 kWh/m² year. For WWR 50, the energy performance varies by +107.57 %; the sDA, 43.33 %, and the UDI, 0 % (both with 100 %), with a difference of 1135.96 kWh year or 20.29 kWh/m² year and for WWR 60, the energy performance varies by 102.64 %; the sDA, 30.00 %, and the UDI, 0 % (both with 100 %), with a difference of 1238.87 kWh year or 22.12 kWh/m² year.

For all the WWRs considered, the best energy performance values have a variation of up to 18 kWh/m² year, and, for the light performances, this variation reaches up to 5 kWh/m² year. The differences are greater between the optimal performance by WWR, exceeding in two of the three WWR levels, twice the lower value, associated with energy performance. For WWR40, the optimal thermal performance is associated with an MSF of 0.36 (Overhang of 170 cm), and for WWR50–60, an MSF of 0.14 (Overhang of 400 cm). Optimal lighting performance for WWR40–60 has an MSF of 0.51 (Overhang of 80 and 170 cm) and for WWR50, has an MSF of 0.36 (Overhang of 160 cm).

4.4. West orientation

For WWR 40, the energy performance varies by +109.30 %; the sDA, 43.84 %, and the UDI, 0 % (both with 100 %), with a difference of 1278.27 kWh year or 22.82 kWh/m² year. For WWR 50, the energy performance varies by +133.53 %; the sDA, 43.67 %, and the UDI, 0 % (both with 100 %), with a difference of 1261.41 kWh year or 20.52 kWh/m² year, and for WWR 60, the energy performance varies by +71.16 %, the sDA, 19.00 %, and the UDI, 0 % (both with 100 %), with a difference of 789.62 kWh year or 14.10 kWh/m² year.

For all the WWRs considered, the best energy performance values have a variation of up to 4 kWh/m² year, and between the light performances, this variation reaches 10 kWh/m² year. The differences are greater between the optimal performance by WWR, exceeding in two of the three WWR levels, twice the lower value, associated with energy performance. It is seen that for WWR40–50 and 60, optimal thermal performance is associated with an MSF of 0.14 (Overhang of 245 cm and 400 cm). Optimal lighting performance for WWR40 has an MSF of 0.51 (Overhang of 90) and for WWR50–60 has an MSF of 0.36 (Overhangs of 160 cm and 245 cm).

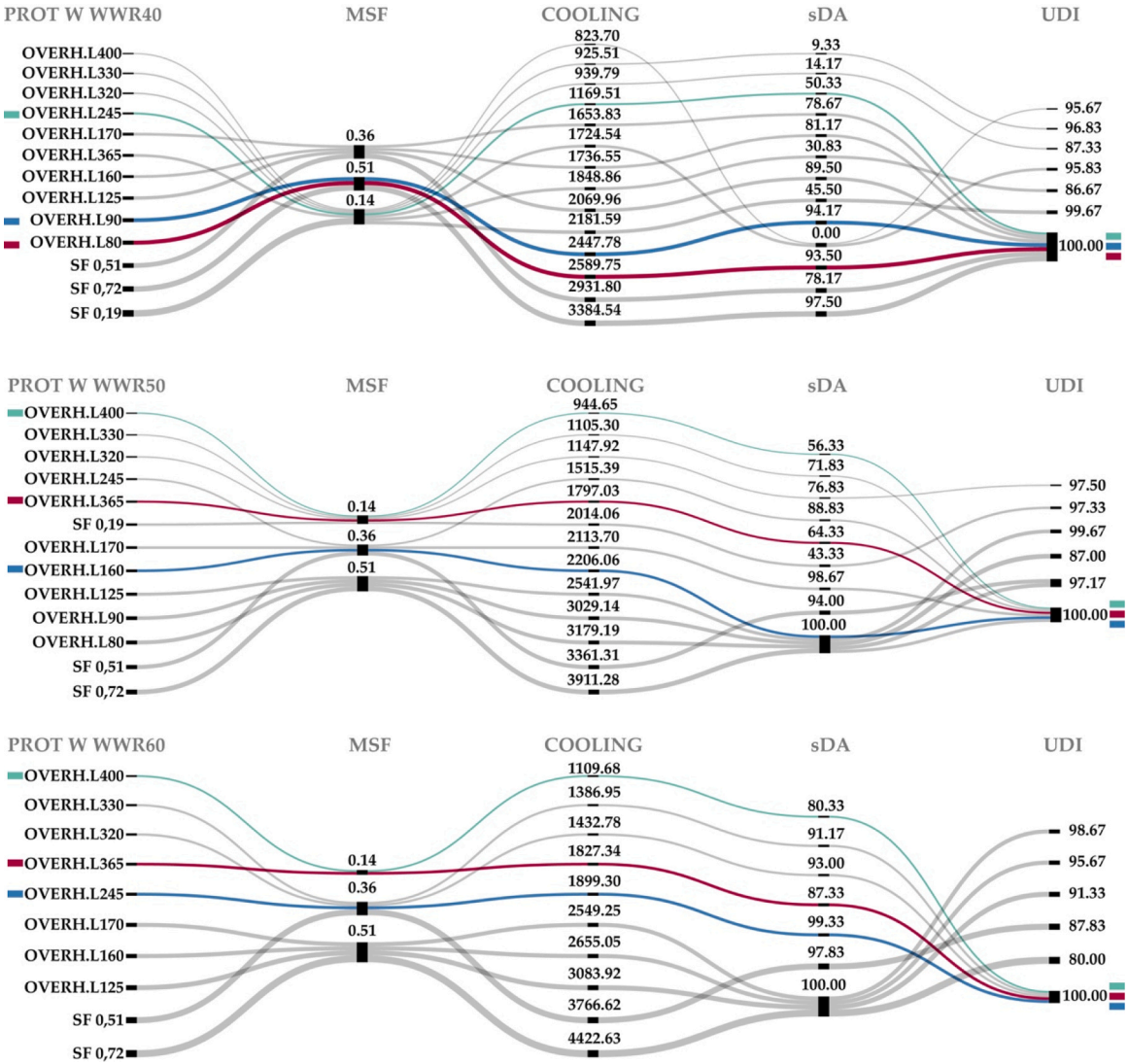


Fig. 16. Characteristic values of the West facing solutions with WWR40–50 and 60 and optimal solutions for the cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI.

Table 8
40th percentile values for the parameters of the different WWRs applied to the Northeast and Northwest orientations.

NORTHEAST NORTHWEST WWR	PERCENTIL	REF kWh año	REF_m² kWh/m² año	REFporc_ah %	sDA %	UDI %
40 NE	40	1232,36	22,00	69,39	58,30	100,00
50 NE	40	1591,73	31,79	62,60	87,43	100,00
60 NE	40	1443,27	25,77	73,50	97,33	91,33
40 NO	40	1198,72	21,40	68,83	58,37	99,97
50 NO	40	1568,16	28,00	65,15	88,20	99,97
60 NO	40	1443,37	25,77	71,83	98,17	93,00

WWR: Window-to-wall ratio; REF: Energy consumption for cooling REF_m2: energy consumption for cooling per square meter; REFporc_ah: percentage of cooling energy savings based on the highest value in the series; sDA: spatial light autonomy; UDI: Useful daylight illuminance.

4.5. Northeast orientation

For WWR 40, the energy performance varies by +164.17 %; the sDA, 22.00 %, and the UDI, 0 % (both with 100 %), with a difference of 1573.91 kWh year or 28.10 kWh/m² year. For WWR 50, the energy performance varies by +110.64 %; the sDA, 29.33 %, and the UDI, 0 % (both with 100 %), with a difference of 946.66 kWh year or 16.90 kWh/m² year, and for WWR 60, the energy performance varies by 59.54 %;

the sDA, 10.00 %, and the UDI, 0 % (both with 100 %), with a difference of 538.63 kWh year or 9.62 kWh/m² year.

For all the WWRs considered, the best energy performance values have a variation of up to 1.84 kWh/m² year, and, for the light performances, this variation reaches up to 19.45 kWh/m² year. The differences are greater between the optimal performance by WWR, exceeding in two of the three WWR levels, twice the lower value, associated with energy performance. For WWR40–50 and 60, the optimum thermal

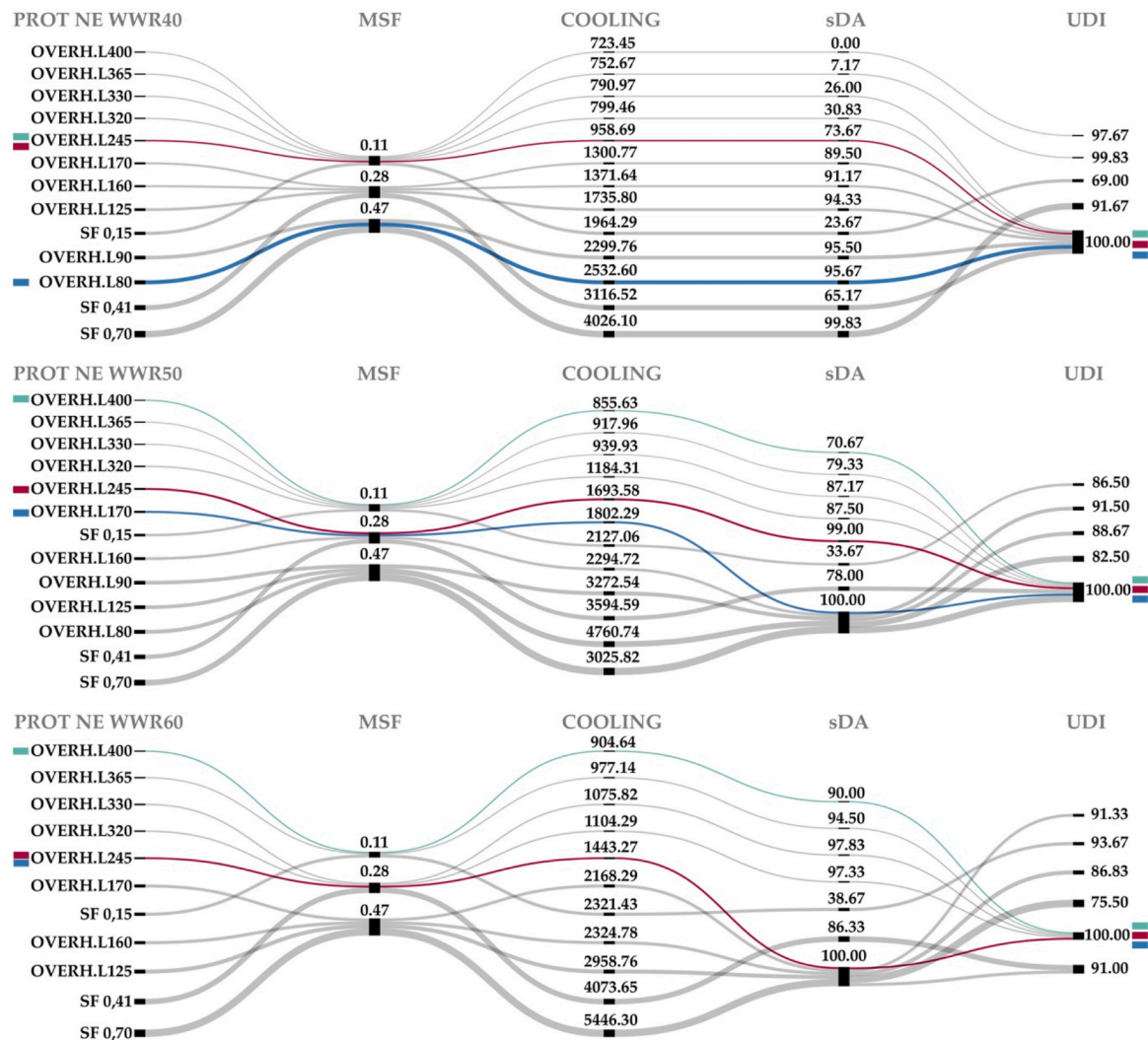


Fig. 17. Characteristic values of the Northeast facing solutions with WWR40–50 and 60 and optimal solutions for the cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI.

performance is associated with an MSF of 0.11 (Overhangs of 245 cm and 400 cm). Optimal lighting performance for WWR40 has an MSF of 0.47 (Overhang of 80 cm) and for WWR50–60 has an MSF of 0.28 (Overhangs of 170 cm and 245 cm).

4.6. Northwest orientation

For WWR 40, the energy performance varies by +165.46 %; the sDA, 21.50 %, and the UDI, 0 % (both with 100 %), with a difference of 1502.71 kWh year or 26.82 kWh/m² year. For WWR 50, the energy performance varies by +116.77 %; the sDA, 26.50 %, and the UDI, 0 % (both with 100 %), with a difference of 898.50 kWh year or 16.04 kWh/m² year, and for WWR 60, the energy performance varies by +67.00 %; the sDA, 8.83 %, and the UDI, 0 % (both with 100 %), with a difference of 579.08 kWh year or 10.34 kWh/m² year.

For all the WWRs considered, the best energy performance values have a variation of up to 2.47 kWh/m² year, and, for the light performances, this variation reaches up to 17.26 kWh/m² year. The differences are greater between the optimal performance by WWR, exceeding in two of the three WWR levels, twice the lower value, associated with energy performance. For WWR40–50 and 60 the optimum thermal performance is associated with an MSF of 0.11 (Overhangs of 245 cm and 400 cm). Optimal lighting performance for WWR40 has an MSF of 0.47 (Overhang of 80 cm) and for WWR50–60 has an MSF of 0.28

(Overhangs of 170 cm and 245 cm).

4.7. General considerations

The analysis is done under a "shoebox" type model, although with characteristics closer to a classroom typology, with WWR of 40 %, 50 %, and 60 %, taking into account that these preliminary models allow initial approaches, and do not always represent reality (Ayoub, 2019).

It has been possible to show that sDA and UDI are suitable as indicators of daylight contributions (Košir et al., 2018; Sun et al., 2020; Turan et al., 2020; Zomorodian and Tahsildoust, 2019) although sDA may comply, ASE does not (Englezou and Michael, 2020). In turn, the ASE analyses performed showed values that were always far from the recommended values without managing to approach the acceptable reference. Therefore, it is suggested not to consider ASE for making early design decisions, as the other two parameters can provide greater clarity regarding daylight minimums or excesses.

Although the results do not show great differences between the E-W and NE-NW pairs, it is considered appropriate to evaluate each case individually since certain combinations may present relevant end energy consumptions.

Even though the 40th percentile is a recommended reference, it was found that certain data series do not provide enough options that reach an acceptable level in the variables and may not achieve an optimal

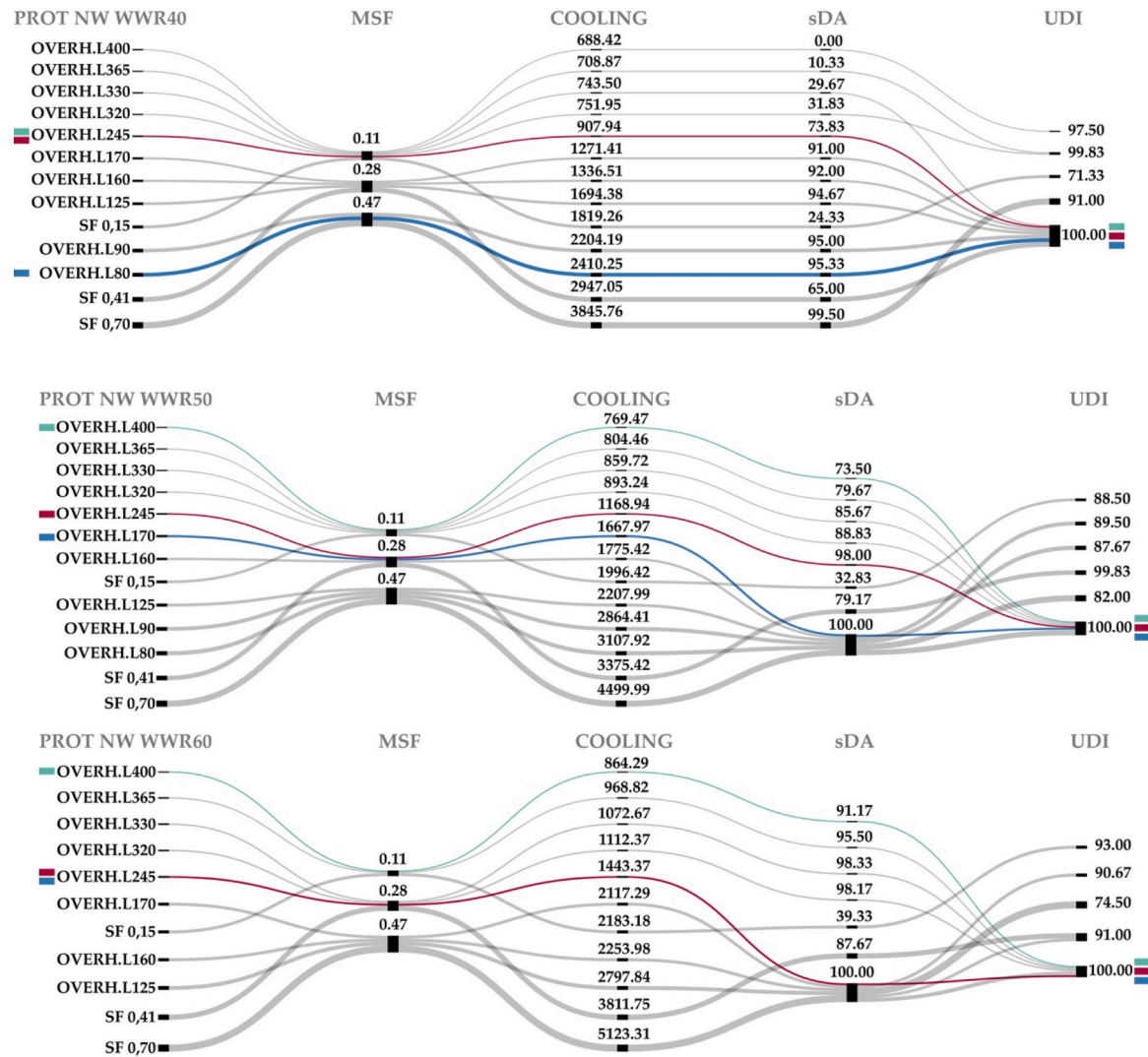


Fig. 18. Characteristic values of the Northwest facing solutions with WWR40–50 and 60 and optimal solutions for the cooling energy consumption parameters in kWh/m² year; percentage energy savings, sDA, and UDI.

solution. However, considering the alternative lower limit defined by the best light performance when the sDA and UDI variables reach the maximum, may expand the battery of solutions. Despite this, it is seen that opting for the best lighting performance can increase energy consumption by twice or more compared to the thermal optimum of a series.

The analysis of the results raises the discussion about the current values associated with MSF since it focuses to a greater extent on thermal performances, leaving aside the light performances relationship. The dynamic analysis allows confirming that solutions under the same MSF value do not necessarily deliver similar light performance values. Although some cases do not have significant differences in this aspect, they do in terms of energy consumption, which can be more than double in some cases.

The following points are considered limitations of this study:

1. The evaluation has considered three WWR proportions (40–50 and 60 %) for school classroom use in a single location in Chile.
2. The analysis was established in an annual period, from January to December, without discounting the summer and winter vacation periods, which are critical periods for energy demand, both in heating/cooling and in daylighting, which would partially reflect the condition of a classroom.

3. Premises with a single exterior facade of reinforced concrete and exterior insulation were considered, without finishes on the interior walls.
4. The weather file was obtained from the Meteonorm v 7.0 software, being able, in some cases, to extrapolate values from places that do not have the same local climatic variations.
5. Glazed surfaces have been defined as single glazing from the Design Builder database, so it does not represent real glazing in all its dimensions. The rest of the glazing comes from the LBNL Window software database and is associated with a real product.
6. The results for ASE showed values above the recommended value for all cases, so it was suggested not to consider this parameter for early design evaluations. However, it is possible that with other strategies or combinations of elements, it is possible to achieve recommended values.
7. Protection with overhangs and optimized windows have been considered, however, it is felt appropriate to evaluate other recommended solar protection options for E-W orientations either through horizontal or vertical louvers.

While the article may have the potential to promote broader applications and universal adoption, the aim of this manuscript is to define a methodology that delves into solar protection solutions by considering

Table 9

Results for the best performances by WWR and by orientation.

WWR NORTH ORIENTATION								
WWR	Category	PROT	MSF	REF kWh year	REF_m ² kWh/m ² year	REFporc_ah %	sDA %	UDI %
40	Best energy performance	Overhang L245	0.11	680.43	12.13	82.97	77.00	100.00
40	Best lighting performance	Overhang L080	0.33	1543.58	27.51	61.36	98.50	100.00
50	Best energy performance	Overhang L400	0.11	646.48	11.52	86.49	70.67	100.00
50	Best lighting performance	Overhang L170	0.18	1135.44	20.24	76.27	100.00	100.00
60	Best energy performance	Overhang L400	0.11	693.95	12.37	87.41	93.17	100.00
60	Best lighting performance	Overhang L245	0.18	1089.09	19.41	80.24	100.00	100.00
WWR EAST ORIENTATION								
WWR	Category	PROT	MSF	REF kWh year	REF_m ² kWh/m ² year	REFporc_ah %	sDA %	UDI %
40	Best energy performance	Overhang L170	0.36	1724.54	30.79	47.11	78.17	100.00
40	Best lighting performance	Overhang L080	0.51	2489.67	44.45	23.64	93.67	100.00
50	Best energy performance	Overhang L400	0.14	1056.03	18.85	72.21	52.67	100.00
50	Best lighting performance	Overhang L160	0.36	2191.99	39.14	42.32	100.00	100.00
60	Best energy performance	Overhang L400	0.14	1207.03	21.55	72.11	70.00	100.00
60	Best lighting performance	Overhang L170	0.51	2445.90	43.67	43.48	100.00	100.00
WWR WEST ORIENTATION								
WWR	Category	PROT	MSF	REF kWh year	REF_m ² kWh/m ² year	REFporc_ah %	sDA %	UDI %
40	Best energy performance	Overhang L245	0.14	1169.51	20.88	65.45	50.33	100.00
40	Best lighting performance	Overhang L090	0.51	2447.78	43.70	27.68	94.17	100.00
50	Best energy performance	Overhang L400	0.14	944.65	16.87	75.85	56.33	100.00
50	Best lighting performance	Overhang L160	0.36	2206.06	39.39	43.60	100.00	100.00
60	Best energy performance	Overhang L400	0.14	1109.68	19.81	74.91	80.33	100.00
60	Best lighting performance	Overhang L245	0.36	1899.30	33.91	57.05	99.33	100.00
WWR NORTHEAST ORIENTATION								
WWR	Category	PROT	FSM	REF kWh año	REF_m ² kWh/m ² año	REFporc_ah %	sDA %	UDI %
40	Better energy performance	Overhang L245	0,11	958,69	17,12	76,19	73,67	100,00
40	Better lighting performance	Overhang L080	0,47	2532,60	45,22	37,10	95,67	100,00
50	Better energy performance	Overhang L400	0,11	813,48	14,52	82,91	70,67	100,00
50	Better lighting performance	Overhang L170	0,28	1693,58	30,24	64,43	100,00	100,00
60	Better energy performance	Overhang L400	0,11	904,64	16,15	83,39	90,00	100,00
60	Better lighting performance	Overhang L245	0,28	1443,27	25,77	73,50	100,00	100,00
WWR NORTHWEST ORIENTATION								
WWR	Category	PROT	MSF	REF kWh year	REF_m ² kWh/m ² year	REFporc_ah %	sDA %	UDI %
40	Best energy performance	Overhang L245	0.11	907.94	16.21	76.39	73.83	100.00
40	Best lighting performance	Overhang L080	0.47	2410.25	43.03	37.33	95.33	100.00
50	Best energy performance	Overhang L400	0.11	769.47	13.74	82.90	73.50	100.00
50	Best lighting performance	Overhang L170	0.28	1667.97	29.78	62.93	100.00	100.00
60	Best energy performance	Overhang L400	0.11	864.29	15.43	83.13	91.17	100.00
60	Best lighting performance	Overhang L245	0.28	1443.37	25.77	71.83	100.00	100.00

the optimal performance of natural lighting through annual dynamic indicators at the early design stage, while maintaining adequate levels of energy savings. For this purpose, only a local climate has been employed, which may pose a limitation. Nevertheless, it is expected that other researchers will use the described methodology and extend the results to other parts of the world

5. Conclusions

The experiment required 36 thermal energy simulations and 36 light performance simulations for the overhang lengths and lighting with the overhang lengths in the five defined orientations (N, NE, NW, E, and W), plus 20 simulations for the values corresponding to the setback in both performances, using the same orientations. The results generated data and possible graphs for comparison, establishing a methodology that allows looking closer at, during an early design stage, solar protection solutions considering the optimal daylighting performance through annual dynamic indicators while maintaining adequate energy-saving levels.

To be able to link the different solutions, it was necessary to initially group those that could be similar regardless of the WWR they had. This generated two grouping options: by MSF and by WWR. Grouping by MSF allows determining which combination and WWR can achieve a balance between both requirements. However, grouping by WWR allows a closer approximation to the early design decision process because it delivers MSF options based on a possible facade appearance and not the other way around.

The findings showed that the same MSF value generates a series of possible solutions to evaluate both its thermal and light performance by grouping by different WWR. From these results it is possible to observe that the dispersion of the values can generate very efficient solutions from the energy point of view, but not so from the light point of view. The optimal lighting system is presented in most cases as a solution that consumes more cooling energy. It also establishes intermediate options that do not have much difference compared to the optimal lighting case, but do for energy consumption. This procedure allowed grouping these optimal solutions by WWR, with this form being more appropriate for the early design process by first establishing the restriction of the facade configuration over the restriction of solar input. At the same time, the results suggest that it is appropriate to consider sDA and UDI as annual dynamic lighting indicators and discard the use of ASE when a tendency to maintain values outside the acceptable range is repeated in the analyses, thereby managing to establish solutions that deliver a better lighting performance and evaluate these together with the energy performances. The simultaneous evaluation of both thermal and light parameters has allowed raising a discussion on the MSF values associated with regulatory documents in Chile and the impact that certain strategies have on light performance. As noted above, this can have a relevant impact on both the circadian cycle and the users' emotional and psychological aspects. Despite this, the experiment is limited when considering a specific case and climate. Other analyses can be made in the future in different localities, considering alternative uses to the one used, to check the regulatory values that currently apply in other climates and latitudes, to evaluate other solar protection options and

establish the impact they have on lighting performance.

Author Statement

The authors declare that the manuscript has not been produced using AI.

CRediT authorship contribution statement

Alexis Pérez-Fargallo: Writing – review & editing, Supervision, Project administration, Methodology. **Cristian Muñoz-Viveros:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carlos Rubio-Bellido:** Writing – review & editing, Supervision, Project administration, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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